



UANL

UNIVERSIDAD AUTONOMA DE NUEVO LEON



FCFM

FACULTAD DE CIENCIAS FISICO MATEMATICAS

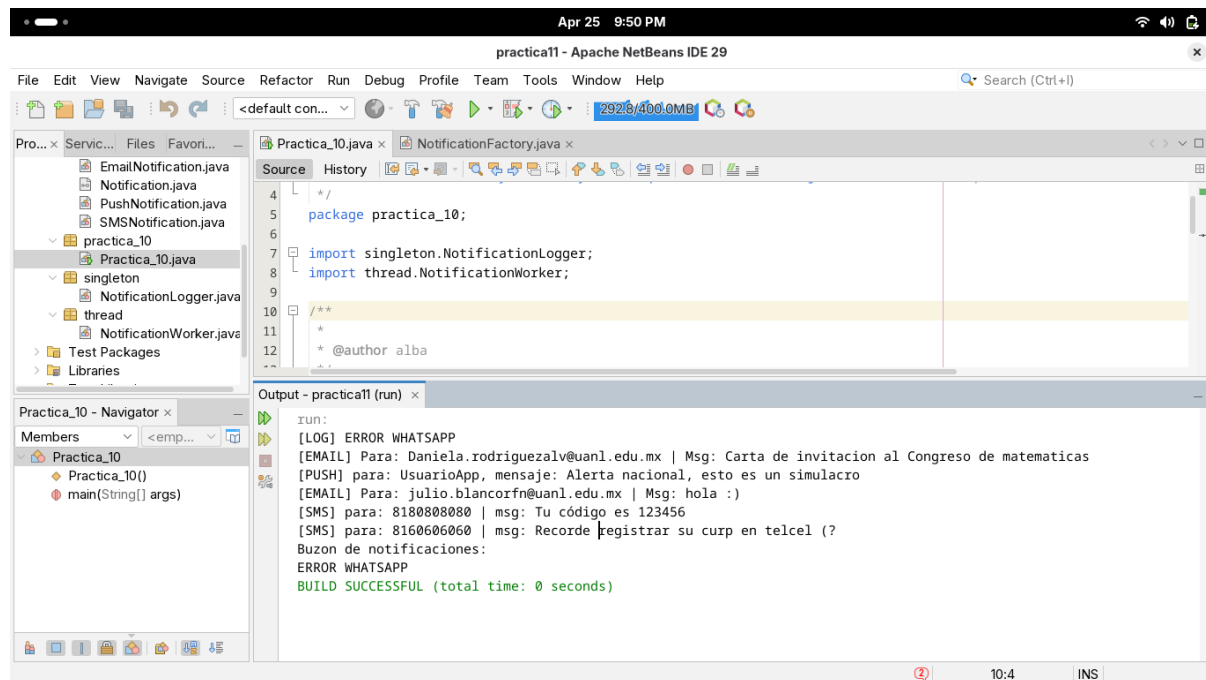
Laboratorio de programacion orientado a objetos Practica 11

Alumno: Julio Antonio Blanco Rufino.

Matricula: 2003632.

Profesora: Jorge Alberto Islas Pineda.

Output:



Main:

/*

- Click nbfs://nbhost/SystemFileSystem/Templates/Licenses/license-default.txt to change this license
- Click nbfs://nbhost/SystemFileSystem/Templates/Classes/Main.java to edit this template

*/ package practica_10;

import singleton.NotificationLogger; import thread.NotificationWorker;

/** *

- @author alba

*/ public class Practica_10 {

public static void main(String[] args) throws InterruptedException {

Thread[] hilos = new Thread[6];

hilos[0] = new Thread(new NotificationWorker("EMAIL",
"julio.blancorfn@uanl.edu.mx", "hola :));

hilos[1] = new Thread(new NotificationWorker("SMS", "8180808080", "Tu código es

```

123456"));
    hilos[2] = new Thread(new NotificationWorker("PUSH", "UsuarioApp", "Alerta
nacional, esto es un simulacro"));
    hilos[3] = new Thread(new NotificationWorker("EMAIL",
"Daniela.rodriquezalv@uanl.edu.mx", "Carta de invitacion al Congreso de
maticas"));
    hilos[4] = new Thread(new NotificationWorker("SMS", "8160606060", "Recorde
registrar su curp en telcel (?"));
    hilos[5] = new Thread(new NotificationWorker("WHATSAPP", "8155555555", "Esto es
invalido"));

    for (Thread hilo : hilos) {
        hilo.start();
    }

    for (Thread hilo : hilos) {
        hilo.join();
    }

    System.out.println("Buzon de notificaciones: ");

    NotificationLogger.getNotificationLogger().getBuzonDeMensajes().forEach(System.out:
:println);
}

}

```

Notification:

```
/*
```

- Click nbfs://nbhost/SystemFileSystem/Templates/Licenses/license-default.txt to change this license
- Click nbfs://nbhost/SystemFileSystem/Templates/Classes/Interface.java to edit this template

```
*/ package model;
```

```
/** *
```

- @author alba

```
*/ public interface Notification { void send(String destinatario); String getType(); }
```

Email:

```
/*
```

- Click nbfs://nbhost/SystemFileSystem/Templates/Licenses/license-default.txt to change this license
- Click nbfs://nbhost/SystemFileSystem/Templates/Classes/Class.java to edit this template

```
*/ package model;
```

```
/** *
```

- @author alba

```
*/ public class EmailNotification implements Notification{ private String mensaje;
```

```
public EmailNotification(String mensaje){  
    this.mensaje = mensaje;  
}
```

```
@Override
```

```
public void send(String destinatario) {  
    System.out.println("[EMAIL] Para: " + destinatario + " | Msg: " +  
        mensaje);  
}
```

```
@Override
```

```
public String getType() {  
    return "EMAIL";  
}
```

```
}
```

Push notification:

```
/*
```

- Click nbfs://nbhost/SystemFileSystem/Templates/Licenses/license-default.txt to change this license

- Click nbfs://nbhost/SystemFileSystem/Templates/Classes/Class.java to edit this template

```
*/ package model;
```

```
/** *
```

- @author alba

```
*/ public class PushNotification implements Notification{ private String mensaje;
```

```
public PushNotification(String mensaje){
    this.mensaje = mensaje;
}
```

```
@Override
```

```
public void send(String destinatario) {
    System.out.println("[PUSH] para: "+destinatario + ", mensaje: " +
        mensaje);
}
```

```
@Override
```

```
public String getType() {
    return "PUSH";
}
```

```
}
```

Sms:

```
/*
```

- Click nbfs://nbhost/SystemFileSystem/Templates/Licenses/license-default.txt to change this license
- Click nbfs://nbhost/SystemFileSystem/Templates/Classes/Class.java to edit this template

```
*/ package model;
```

```
/** *
```

- @author alba

```
*/ public class SMSNotification implements Notification{ private String mensaje;
```

```

public SMSNotification(String mensaje){
    this.mensaje = mensaje;
}

@Override
public void send(String destinatario) {
    System.out.println("[SMS] para: " + destinatario + " | msg: " +
        mensaje);
}

@Override
public String getType() {
    return "SMS";
}

}

```

Factory:

```

/*
    • Click nbfs://nbhost/SystemFileSystem/Templates/Licenses/license-default.txt
      to change this license
    • Click nbfs://nbhost/SystemFileSystem/Templates/Classes/Class.java to edit
      this template

*/ package factory;

import excepciones.InvalidNotificationTypeException; import model.EmailNotification;
import model.Notification; import model.PushNotification; import
model.SMSNotification;

/**
    • @author alba

*/ public class NotificationFactory {

public static Notification create(String tipo, String mensaje) throws
InvalidNotificationTypeException{
    if(tipo == null){
        throw new InvalidNotificationTypeException("null");
    }
}
}

```

```

    }

    switch(tipo.toUpperCase()){
        case "EMAIL":
            return new EmailNotification(mensaje);
        case "SMS":
            return new SMSNotification(mensaje);
        case "PUSH":
            return new PushNotification(mensaje);
        default:
            throw new InvalidNotificationTypeException(tipo);
    }
}

```

```

}

```

Singleton:

```

/*

```

- Click nbfs://nbhost/SystemFileSystem/Templates/Licenses/license-default.txt to change this license
- Click nbfs://nbhost/SystemFileSystem/Templates/Classes/Class.java to edit this template

```

*/ package singleton;

```

```

import java.util.ArrayList; import java.util.List; /** *

```

- @author alba

```

*/ public class NotificationLogger {

```

```

    private static NotificationLogger notificaciones;
    private List<String> buzonDeMensajes;

```

```

    private NotificationLogger(){
        buzonDeMensajes = new ArrayList<>();
    }

```

```

    public static synchronized NotificationLogger getNotificationLogger(){
        if(notificaciones == null){

```

```

        notificaciones = new NotificationLogger();
    }
    return notificaciones;
}

public synchronized void ingresarBuzon(String mensaje){
    buzonDeMensajes.add(mensaje);
    System.out.println("[LOG] " + mensaje);
}

public synchronized List<String> getBuzonDeMensajes(){
    return new ArrayList<>(buzonDeMensajes);
}

}

```

Thread:

```
/*
```

- Click nbfs://nbhost/SystemFileSystem/Templates/Licenses/license-default.txt to change this license
- Click nbfs://nbhost/SystemFileSystem/Templates/Classes/Class.java to edit this template

```
*/ package thread;
```

```
import factory.NotificationFactory; import model.Notification; import
singleton.NotificationLogger; import excepciones.InvalidNotificationTypeException;
```

```
/** *
```

- @author alba

```
*/ public class NotificationWorker implements Runnable{ private String tipo; private
String destinatario; private String mensaje;
```

```
public NotificationWorker(String tipo, String destinatario, String mensaje){
    this.tipo = tipo;
    this.destinatario = destinatario;
    this.mensaje = mensaje;
}

```

```

@Override
public void run() {
    try{
        Notification notificacion = NotificationFactory.create(tipo, mensaje);

        notificacion.send(destinatario);

        String registrarMensaje = "Enviada" + notificacion.getType() + " a " +
            destinatario;
    }
    catch(InvalidNotificationTypeException e){
        NotificationLogger.getNotificationLogger().ingresarBuzon("ERROR
"+e.getMessage());
    }
}

}

```

Exception:

```
/*
```

- Click nbfs://nbhost/SystemFileSystem/Templates/Licenses/license-default.txt to change this license
- Click nbfs://nbhost/SystemFileSystem/Templates/Classes/Class.java to edit this template

```
*/ package excepciones;
```

```
/** *
```

- @author alba

```
*/ public class InvalidNotificationTypeException extends Exception{ public
InvalidNotificationTypeException(String message) { super(message); } }
```